

Задача 8 Первые интегралы. Скобки Пуассона

220.17

Дано: $\varphi_1 = p_1^2 + q_2^2$, $\varphi_2 = p_2^2 + q_1^2$, $\varphi_3 = (\varphi_1, \varphi_2)$, $H = p_1 p_2 + q_1 q_2$

Показать, что φ_1, φ_2 и φ_3 — независимые первые интегралы

Решение:

Критерий первого интеграла: $\frac{\partial \varphi}{\partial t} + (\varphi, H) = 0$

$$\frac{\partial \varphi_1}{\partial t} = 0; \frac{\partial \varphi_1}{\partial \vec{q}} = \|0 \quad 2q_2\|, \frac{\partial \varphi_1}{\partial \vec{p}} = \|2p_1 \quad 0\|, \frac{\partial \varphi_2}{\partial \vec{q}} = \|2q_1 \quad 0\|, \frac{\partial \varphi_2}{\partial \vec{p}} = \|0 \quad 2p_2\|$$

$$\frac{\partial H}{\partial \vec{q}} = \|q_2 \quad q_1\|^T, \frac{\partial H}{\partial \vec{p}} = \|p_2 \quad p_1\|^T$$

$$1) (\varphi_1, H) = \frac{\partial \varphi_1}{\partial \vec{q}} \frac{\partial H}{\partial \vec{p}} - \frac{\partial \varphi_1}{\partial \vec{p}} \frac{\partial H}{\partial \vec{q}} = 2q_2 p_1 - 2q_2 p_1 = 0 \Rightarrow \varphi_1 - \text{первый интеграл}$$

$$2) (\varphi_2, H) = \frac{\partial \varphi_2}{\partial \vec{q}} \frac{\partial H}{\partial \vec{p}} - \frac{\partial \varphi_2}{\partial \vec{p}} \frac{\partial H}{\partial \vec{q}} = 2q_1 p_2 - 2p_2 q_1 = 0 \Rightarrow \varphi_2 - \text{первый интеграл}$$

$$3) \varphi_3 = \frac{\partial \varphi_1}{\partial \vec{q}} \frac{\partial \varphi_2}{\partial \vec{p}} - \frac{\partial \varphi_1}{\partial \vec{p}} \frac{\partial \varphi_2}{\partial \vec{q}} = 4p_2 q_2 - 4p_1 q_1$$

$$\frac{\partial \varphi_3}{\partial \vec{q}} = \|-4p_1 \quad 4p_2\|, \frac{\partial \varphi_3}{\partial \vec{p}} = \|-4q_1 \quad 4q_2\|$$

$$(\varphi_3, H) = \frac{\partial \varphi_3}{\partial \vec{q}} \frac{\partial H}{\partial \vec{p}} - \frac{\partial \varphi_3}{\partial \vec{p}} \frac{\partial H}{\partial \vec{q}} = (-4p_1 p_2 + 4p_2 p_1) - (-4q_1 q_2 + 4q_2 q_1) = 0 \Rightarrow$$

$\Rightarrow \varphi_3$ — первый интеграл по критерию.

Проверим функциональную независимость с помощью Якобиана:

$$\begin{vmatrix} 0 & 2q_2 & 2p_1 & 0 \\ 2q_1 & 0 & 0 & 2p_2 \\ -4p_1 & 4p_2 & -4q_1 & 4q_2 \end{vmatrix} \sim \begin{vmatrix} 1 & 0 & p_1 & p_2 \\ 0 & 1 & q_1 & q_2 \\ 1 & -\frac{p_2}{p_1} & -\frac{q_2}{p_1} & \frac{q_2}{p_1} \end{vmatrix} \sim \begin{vmatrix} 1 & 0 & p_1 & p_2 \\ 0 & 1 & q_1 & q_2 \\ 0 & -\frac{p_2}{p_1} & -\frac{q_2}{p_1} & \frac{q_2}{p_1} - \frac{p_2}{q_1} \end{vmatrix} \Rightarrow$$

$\Rightarrow \text{rg} = 3$, φ_1, φ_2 и φ_3 — функционально независимые первые интегралы

220.60

Дано: $f_i(\vec{q}, \vec{p}, t) = d_i$, $i \in \overline{1, m}$ — первые интегралы, находящиеся в инволюции, то есть $(f_i, f_j) = 0$

Показать: $F(f_1, \dots, f_m)$ и $\Phi(f_1, \dots, f_m)$ находятся в инволюции

Решение:

$$\begin{aligned} (F, \Phi) &= \sum_{i=1}^n \left(\frac{\partial F}{\partial q_i} \frac{\partial \Phi}{\partial p_i} - \frac{\partial F}{\partial p_i} \frac{\partial \Phi}{\partial q_i} \right) = \sum_{i=1}^n \left[\left(\sum_{j=1}^m \frac{\partial F}{\partial f_j} \frac{\partial f_j}{\partial q_i} \right) \left(\sum_{k=1}^m \frac{\partial \Phi}{\partial f_k} \frac{\partial f_k}{\partial p_i} \right) - \left(\sum_{j=1}^m \frac{\partial F}{\partial f_j} \frac{\partial f_j}{\partial p_i} \right) \cdot \left(\sum_{k=1}^m \frac{\partial \Phi}{\partial f_k} \frac{\partial f_k}{\partial q_i} \right) \right] \\ &= \sum_{i=1}^n \sum_{j,k=1}^m \left(\frac{\partial F}{\partial f_j} \frac{\partial f_j}{\partial q_i} \cdot \frac{\partial \Phi}{\partial f_k} \frac{\partial f_k}{\partial p_i} - \frac{\partial F}{\partial f_j} \frac{\partial f_j}{\partial p_i} \cdot \frac{\partial \Phi}{\partial f_k} \frac{\partial f_k}{\partial q_i} \right) = \\ &= \sum_{j,k=1}^m \frac{\partial F}{\partial f_j} \cdot \frac{\partial \Phi}{\partial f_k} \sum_{i=1}^n \left(\frac{\partial f_j}{\partial q_i} \frac{\partial f_k}{\partial p_i} - \frac{\partial f_j}{\partial p_i} \frac{\partial f_k}{\partial q_i} \right) = \sum_{j,k=1}^m \frac{\partial F}{\partial f_j} \frac{\partial \Phi}{\partial f_k} \cdot (f_j, f_k) = 0 \end{aligned}$$

220.23

Показать, что $f(q_1, \dots, q_m, p_1, \dots, p_m)$ является первым интегралом для системы с гамильтонианом $H(f(q_1, \dots, q_m, p_1, \dots, p_m), q_{m+1}, \dots, q_n, p_{m+1}, \dots, p_n)$

Решение:

$$\frac{\partial f}{\partial q_i} = \left\| \frac{\partial f}{\partial q_1} \dots \frac{\partial f}{\partial q_m} 0 \dots 0 \right\|$$

$$\frac{\partial f}{\partial p_i} = \left\| \frac{\partial f}{\partial p_1} \dots \frac{\partial f}{\partial p_m} 0 \dots 0 \right\|$$

$$\frac{\partial H}{\partial q} = \left\| \frac{\partial H}{\partial f} \frac{\partial f}{\partial q_1} \dots \frac{\partial H}{\partial f} \frac{\partial f}{\partial q_m} \frac{\partial H}{\partial q_{m+1}} \dots \frac{\partial H}{\partial q_n} \right\|^T$$

$$\frac{\partial H}{\partial p} = \left\| \frac{\partial H}{\partial f} \frac{\partial f}{\partial p_1} \dots \frac{\partial H}{\partial f} \frac{\partial f}{\partial p_m} \frac{\partial H}{\partial p_{m+1}} \dots \frac{\partial H}{\partial p_n} \right\|^T$$

$$(f, H) = \frac{\partial f}{\partial q} \frac{\partial H}{\partial p} - \frac{\partial f}{\partial p} \frac{\partial H}{\partial q} = \sum_{i=1}^m \left(\frac{\partial f}{\partial q_i} \frac{\partial H}{\partial f} \frac{\partial f}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial H}{\partial f} \frac{\partial f}{\partial q_i} \right) = 0 \Rightarrow \text{(по критерию первого интеграла: } \frac{\partial f}{\partial t} + (f, H) = 0 \text{) } f \text{ - первый интеграл системы.}$$

220.23(2)

Дано: $H = (p_1^2 + q_1^2) F(p_2, \dots, p_n, t)$

Найти: движение системы

Решение:

$$\dot{q}_1 = \frac{\partial H}{\partial p_1} = 2 p_1 F(p_2, \dots, p_n, t)$$

$$\dot{q}_i = \frac{\partial H}{\partial p_i} = (p_1^2 + q_1^2) \cdot \frac{\partial F}{\partial p_i}(p_2, \dots, p_n, t), i \in \overline{2 \dots n}$$

$$\dot{p}_1 = -\frac{\partial H}{\partial q_1} = -2 q_1 F(p_2, \dots, p_n, t)$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i} = 0, i \in \overline{2 \dots n} \text{ - циклическая переменная}$$

Таким образом:

$$p_i(t) = d_i = \text{const}, i \in \overline{2 \dots n}$$

$$\begin{cases} \dot{q}_1 = 2 p_1 F(d_2, \dots, d_n, t) & (1) \end{cases}$$

$$\begin{cases} \dot{q}_i = (p_1^2 + q_1^2) \frac{\partial F}{\partial p_i}(d_2, \dots, d_n, t) & (2) \end{cases}$$

$$\begin{cases} \dot{p}_1 = -2 q_1 F(d_2, \dots, d_n, t) & (3) \end{cases}$$

$$\frac{d}{dt}(p_1^2 + q_1^2) = 2 p_1 \dot{p}_1 + 2 q_1 \dot{q}_1 = -4 p_1 q_1 F(d_2, \dots, d_n, t) + 4 q_1 p_1 F(d_2, \dots, d_n, t) = 0 \Rightarrow$$

$$\Rightarrow p_1^2 + q_1^2 = d_1 = \text{const}$$

$$i \in \overline{2 \dots n}: q_i(t) = d_1 \int_{t_0}^t \frac{\partial F}{\partial p_i}(d_2, \dots, d_n, t) dt$$

$$(1): \dot{q}_1 = 2 \sqrt{d_1 - q_1^2} F(d_2, \dots, d_n, t)$$

$$\frac{d}{dt} \frac{q_1}{\sqrt{1 - \left(\frac{q_1}{\sqrt{\alpha_1}}\right)^2}} = 2 F(\alpha_2, \dots, \alpha_n, t) dt \Rightarrow \arcsin \frac{q_1}{\sqrt{\alpha_1}} = 2 \int_{t_0}^t F(\alpha_2, \dots, \alpha_n, t) dt \Rightarrow$$

$$\Rightarrow q_1(t) = \sqrt{\alpha_1} \sin \left[\int_{t_0}^t F(\alpha_2, \dots, \alpha_n, t) dt \right]$$

Ответ: $q_1(t) = \sqrt{\alpha_1} \sin \left[\int_{t_0}^t F(\alpha_2, \dots, \alpha_n, t) dt \right], q_i(t) = \alpha_i \int_{t_0}^t \frac{\partial F}{\partial q_i}(\alpha_2, \dots, \alpha_n, t) dt$

20.35

Дано: $L = \frac{1}{2} \sum_{i,j=1}^n a_{ij}(\bar{q}) q_i q_j - \Pi(\bar{q})$, $a_{ij}(\bar{q})$ - однородные порядка k , $\Pi(\bar{q})$ - однородная порядка l , $q_i^* = q_i \exp(\alpha a)$, $t^* = t \exp(\alpha b)$, α - параметр
Найти: связь между k, l, a и b , чтобы выполнялись условия теоремы Гётер, соответствующий первый интеграл.

Решение:

Условия $(q^*, t^*) \equiv (q, t)$ при $\alpha = 0$ и дифференцируемости q^* и t^* по параметру α выполняются. Для выполнения всех условий необходимо, чтобы преобразование не меняло элементарное действие по Гамильтону:

$$dW = L(\bar{q}, \bar{q}, t) dt = \left[\frac{1}{2} \sum_{i,j=1}^n a_{ij}(\bar{q}) \dot{q}_i \dot{q}_j - \Pi(\bar{q}) \right] dt$$

$$dW^* = L(\bar{q}^*, \bar{q}^*, t^*) dt^* = \left[\frac{1}{2} \sum_{i,j=1}^n a_{ij}(e^{\alpha a} \bar{q}) \frac{e^{\alpha a}}{e^{\alpha b}} \dot{q}_i \frac{e^{\alpha a}}{e^{\alpha b}} \dot{q}_j - \Pi(e^{\alpha a} \bar{q}) \right] e^{\alpha b} dt =$$

$$= \left[\frac{1}{2} \sum_{i,j=1}^n e^{(k+\alpha)a - \alpha b} a_{ij}(\bar{q}) q_i q_j - e^{l\alpha a} \Pi(\bar{q}) \right] e^{\alpha b} dt.$$

$dW^* = dW$, когда $(k+\alpha)a - \alpha b = 0$ и $l\alpha a + \alpha b = 0$, то есть теорема Гётер выполняется при $(k+\alpha)a - b = 0$ и $la + b = 0$

По теореме Гётер у системы существует первый интеграл

$$\Phi = \left(\bar{p}^T \frac{\partial \bar{q}^*}{\partial \alpha} - H(\bar{q}, \bar{p}, t) \frac{\partial t^*}{\partial \alpha} \right) \Big|_{\alpha=0}$$

$$\bar{p} = \frac{\partial L}{\partial \dot{\bar{q}}} = A(\bar{q}) \dot{\bar{q}}, \dot{\bar{q}} = A^{-1} \bar{p}$$

$$H(\bar{q}, \bar{p}, t) = \bar{p}^T \dot{\bar{q}} - L = \bar{p}^T A^{-1} \bar{p} - \frac{1}{2} \dot{\bar{q}}^T A \dot{\bar{q}} + \Pi(\bar{q}) = \frac{1}{2} \bar{p}^T A^{-1} \bar{p} + \Pi(\bar{q})$$

$$\frac{\partial \bar{q}}{\partial \alpha} = a \bar{q} e^{\alpha a}, \frac{\partial t^*}{\partial \alpha} = b t e^{\alpha b}$$

$$\Phi(\bar{q}, \bar{p}, t) = a \bar{p}^T \bar{q} - \left(\frac{1}{2} \bar{p}^T A^{-1} \bar{p} + \Pi(\bar{q}) \right) b t = \text{const}$$

Ответ: $(k+\alpha)a - b = 0$ и $la + b = 0$, $a \bar{p}^T \bar{q} - \left(\frac{1}{2} \bar{p}^T A^{-1} \bar{p} + \Pi(\bar{q}) \right) b t = \text{const}$

T3

Дано: Правой части $\dot{\vec{x}} = \vec{f}(\vec{x})$ соответствует оператор дифференцирования $U = f^i \partial_i$ (как меняется h вдоль траектории $\vec{x}(t)$: $\frac{d}{dt} h(\vec{x}(t)) = \sum_{i=1}^n \frac{\partial h}{\partial x_i} \dot{x}_i = \sum_{i=1}^n f^i(\vec{x}) \frac{\partial h}{\partial x_i} = f^i \partial_i$). Коммутатором двух операторов U и V называется оператор $[U, V] = UV - VU = (Ug^i - Vf^i) \partial_i$.

Показать, что коммутатор $[U_F, U_G]$ операторов систем с гамильтонианами F и G - оператор системы с гамильтонианом равным их скобке Пуассона $H = (F, G)$.

Решение:

$$U_F = \sum_{i=1}^{2n} f_i \frac{\partial}{\partial x_i} = \sum_{i=1}^n \frac{\partial F}{\partial p_i} \frac{\partial}{\partial q_i} - \sum_{i=1}^n \frac{\partial F}{\partial q_i} \frac{\partial}{\partial p_i} = (F, \cdot)$$

$$\text{Аналогично } U_G = (G, \cdot), U_H = (H, \cdot) = ((F, G), \cdot)$$

$$[U_F, U_G] = \sum_{i=1}^{2n} (U_F g_i - U_G f_i) \frac{\partial}{\partial x_i} = \sum_{i=1}^n (U_F \frac{\partial G}{\partial p_i} - U_G \frac{\partial F}{\partial p_i}) \frac{\partial}{\partial q_i} - \sum_{i=1}^n (U_F \frac{\partial G}{\partial q_i} - U_G \frac{\partial F}{\partial q_i}) \frac{\partial}{\partial p_i}$$

$$= U_F \left(\frac{\partial G}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial G}{\partial q_i} \frac{\partial}{\partial p_i} \right) - U_G \left(\frac{\partial F}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial F}{\partial q_i} \frac{\partial}{\partial p_i} \right) = U_F U_G - U_G U_F$$

$$= \left(\frac{\partial F}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial F}{\partial q_i} \frac{\partial}{\partial p_i} \right) \left(\frac{\partial G}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial G}{\partial q_i} \frac{\partial}{\partial p_i} \right) - \left(\frac{\partial G}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial G}{\partial q_i} \frac{\partial}{\partial p_i} \right) \left(\frac{\partial F}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial F}{\partial q_i} \frac{\partial}{\partial p_i} \right) =$$

$$= \frac{\partial F}{\partial p_i} \left(\frac{\partial^2 G}{\partial q_i \partial p_i} \frac{\partial}{\partial q_i} + \frac{\partial G}{\partial p_i} \frac{\partial^2}{\partial q_i \partial p_i} - \frac{\partial^2 G}{\partial q_i \partial p_i} \frac{\partial}{\partial p_i} - \frac{\partial G}{\partial q_i} \frac{\partial^2}{\partial q_i \partial p_i} \right) -$$

$$- \frac{\partial F}{\partial q_i} \left(\frac{\partial^2 G}{\partial p_i \partial p_i} \frac{\partial}{\partial q_i} + \frac{\partial G}{\partial p_i} \frac{\partial^2}{\partial p_i \partial q_i} - \frac{\partial^2 G}{\partial p_i \partial q_i} \frac{\partial}{\partial p_i} - \frac{\partial G}{\partial q_i} \frac{\partial^2}{\partial p_i \partial p_i} \right) -$$

$$- \frac{\partial G}{\partial p_i} \left(\frac{\partial^2 F}{\partial q_i \partial p_i} \frac{\partial}{\partial q_i} + \frac{\partial F}{\partial p_i} \frac{\partial^2}{\partial q_i \partial p_i} - \frac{\partial^2 F}{\partial q_i \partial p_i} \frac{\partial}{\partial p_i} - \frac{\partial F}{\partial q_i} \frac{\partial^2}{\partial q_i \partial p_i} \right) +$$

$$+ \frac{\partial G}{\partial q_i} \left(\frac{\partial^2 F}{\partial p_i \partial p_i} \frac{\partial}{\partial q_i} + \frac{\partial F}{\partial p_i} \frac{\partial^2}{\partial p_i \partial q_i} - \frac{\partial^2 F}{\partial p_i \partial q_i} \frac{\partial}{\partial p_i} - \frac{\partial F}{\partial q_i} \frac{\partial^2}{\partial p_i \partial p_i} \right) =$$

$$= \left(\frac{\partial F}{\partial p_i} \frac{\partial^2 G}{\partial q_i \partial p_i} - \frac{\partial F}{\partial q_i} \frac{\partial^2 G}{\partial p_i \partial p_i} - \frac{\partial G}{\partial p_i} \frac{\partial^2 F}{\partial q_i \partial p_i} + \frac{\partial G}{\partial q_i} \frac{\partial^2 F}{\partial p_i \partial p_i} \right) \frac{\partial}{\partial q_i} -$$

$$- \left(\frac{\partial F}{\partial p_i} \frac{\partial^2 G}{\partial q_i \partial q_i} - \frac{\partial F}{\partial q_i} \frac{\partial^2 G}{\partial p_i \partial q_i} - \frac{\partial G}{\partial p_i} \frac{\partial^2 F}{\partial q_i \partial q_i} + \frac{\partial G}{\partial q_i} \frac{\partial^2 F}{\partial p_i \partial q_i} \right) \frac{\partial}{\partial p_i} =$$

$$= \frac{\partial}{\partial p_i} \left(\frac{\partial F}{\partial p_i} \frac{\partial G}{\partial q_i} - \frac{\partial G}{\partial p_i} \frac{\partial F}{\partial q_i} \right) \frac{\partial}{\partial q_i} - \frac{\partial}{\partial q_i} \left(\frac{\partial F}{\partial p_i} \frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial q_i} \frac{\partial G}{\partial q_i} \right) \frac{\partial}{\partial p_i} =$$

$$= ((F, G), \cdot) = U_H$$

Таким образом, $[U_F, U_G] = U_H$